(Fd), Plastoquinone (PQ), cytochrome b6-f complex, Plastocyanin (PC) and finally back to chlorophyll P700 (PS I). During this movement of electrons Adenosine Di Phosphate (ADP) is phosphorylated, by the addition of inorganic phosphate and generates Adenosine Tri Phosphate (ATP). Cyclic electron transport produces only ATP and there is no NADPH + H^+ formation. At each step of electron transport, electron loses potential energy and is used by the transport chain to pump H^+ ions across the thylakoid membrane. The proton gradient triggers ATP formation in ATP synthase enzyme situated on the thylakoid membrane. Photosystem I need light of longer wave length (> P700 nm). It operates under low light intensity, less CO_2 and under anaerobic conditions which makes it considered as earlier in evolution (Figure 13.13).

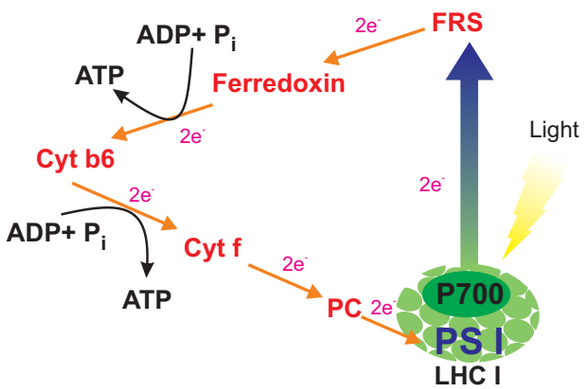


Figure 13.13: Cyclic Photophosphorylation

13.10.2 Non-Cyclic Photophosphorylation

When photons are activated reaction centre of pigment system II(P680), electrons moved to the high energy level. Electrons from high energy state passes through series of electron carriers like pheophytin, plastoquinone, cytochrome complex, plastocyanin and finally accepted by PS I (P700). During this movement of electrons from PS II to PS I ATP is generated (Figure 13. 16). PS I (P700) is activated by light, electrons are moved to high energy state and accepted by electron acceptor molecule ferredoxin reducing Substance (FRS). During the downhill movement through ferredoxin, electrons are transferred to $NADP^+$ and reduced into NADPH + H^+ (H^+ formed from splitting of water by light).

Electrons released from the photosystem II are not cycled back. It is used for the reduction of $NADP^+$ in to NADPH + H^+. During the electron transport it generates ATP and hence this type of photophosphorylation is called **non-cyclic photophosphorylation**. The electron flow looks like the appearance of letter 'Z' and so known as **Z scheme**. When there is availability of $NADP^+$ for reduction and when there is splitting of water molecules both PS I and PS II are activated (Table 13.3). Non-cyclic electron transport PS I and PS II both are involved co-operatively to transport electrons from water to $NADP^+$ (Figure 13.14).

13.10.3 Bio energetics of light reaction

- To release one electron from pigment system it requires two quanta of light.
- One quantum is used for transport of electron from water to PS I.
- Second quantum is used for transport of electron from PS I to $NADP^+$
- Two electrons are required to generate one NADPH + H^+.

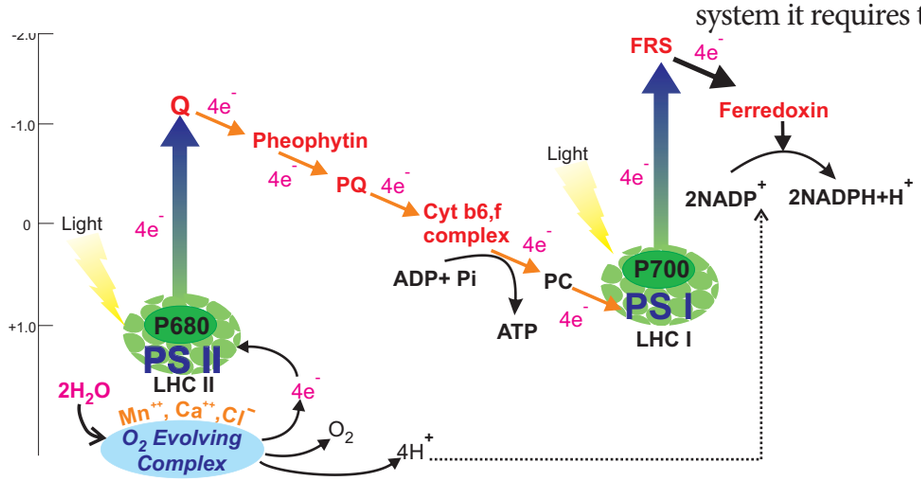


Figure 13.14: Non-Cyclic Photophosphorylation

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